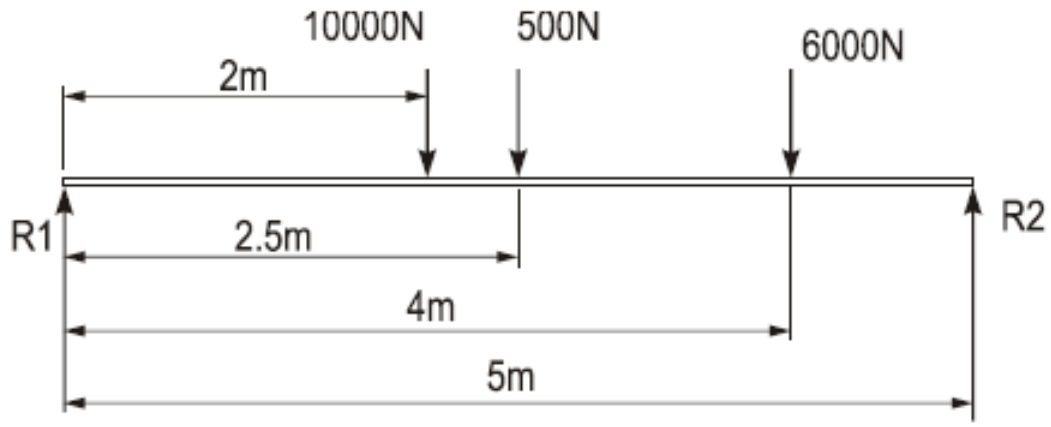
Moments and Reactions on a Structure Worksheet

Worked example

Determine the reactions R_1 and R_2 for the simply supported beam.



Take moments about R_1

$$\Sigma \text{ Clockwise moments} = \Sigma \text{ Anti Clockwise moments}$$

$$(10{,}000\ \text{N} \times 2\ \text{m}) + (500\ \text{N} \times 2.5\ \text{m}) + (6000\ \text{N} \times 4\ \text{m}) = R_2 \times 5\ \text{m}$$

$$R_2 = \frac{20{,}000 + 1250 + 24{,}000}{5}$$

$$\mathbf{R_2 = 9050\ N}$$

Also Σ upwards forces = Σ downwards forces

$$R_1 + 9050 = 10{,}000 + 500 + 6000$$

$$R_1 = 16{,}500 - 9050$$

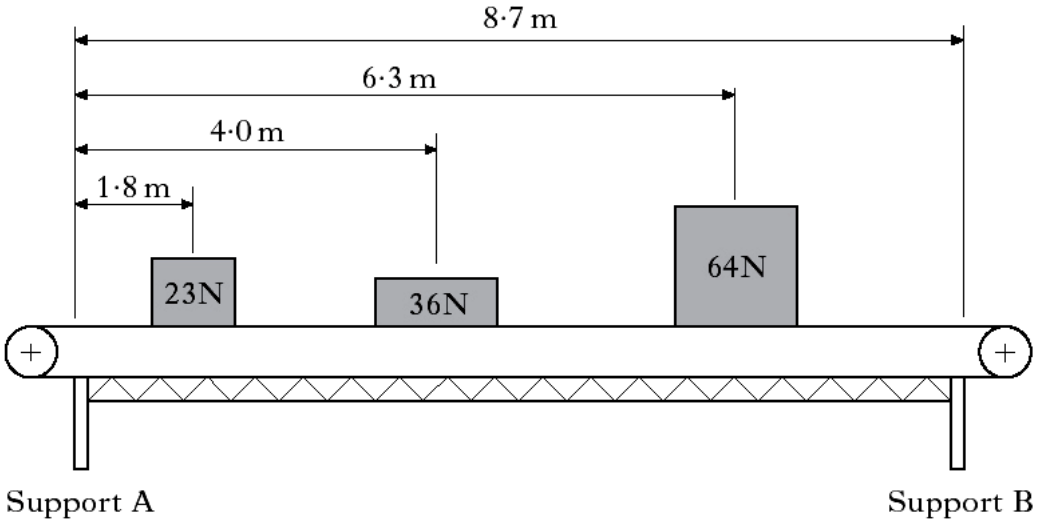
$$\mathbf{R_1 = 7450\ N}$$

Therefore the reactions for the beam supports are

$R_1 = 7450$ N and $R_2 = 9050$ N

In your jotters, draw a free body diagram of the pictures shown below, indicating clearly all forces and reactions.

1. A conveyor belt is used to move packages in a warehouse. When the conveyor belt stopped, packages were in the position shown.



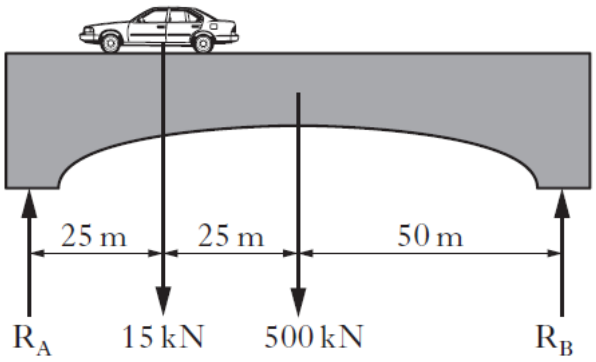
(i) Draw the free body diagram for the system.

(ii) Calculate, by taking moments about support A, the reaction at support B.

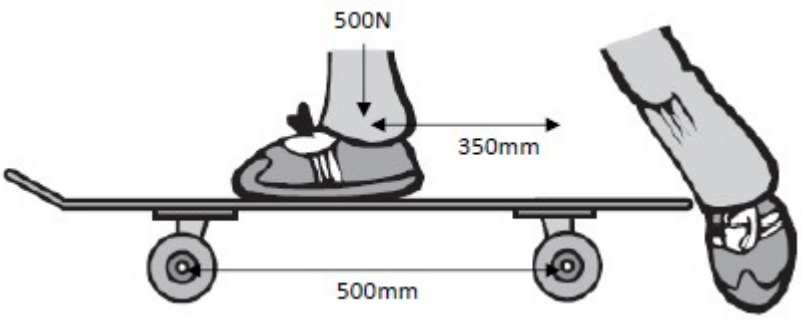
(iii) Determine the reaction at support A.

2. As part of an army training routine, soldiers have to climb across monkey bars 25 metres long. A solider weighing 95kg starts on the left hand side but stops 9 metres along to stop to have a rest. Draw the free body diagram.

3. Draw the free body diagram of the bridge structure shown below.



4. Produce a free body diagram in your jotter and calculate.

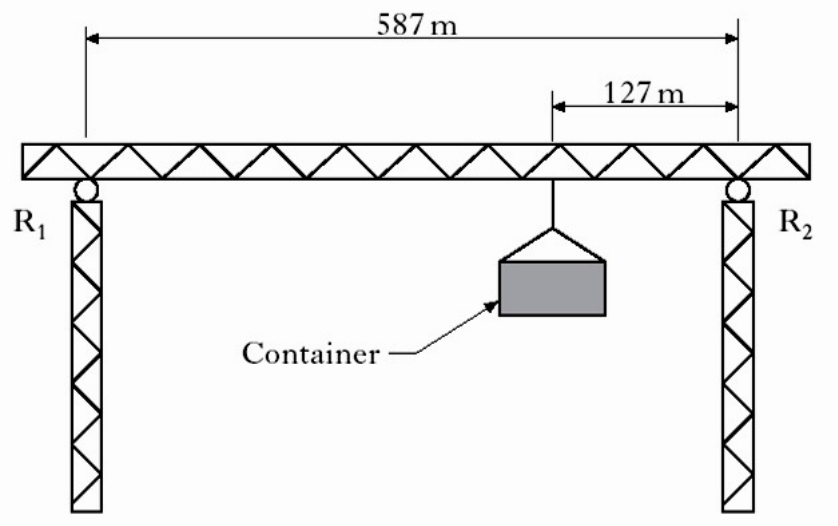


(i) Draw the free body diagram.

(ii) Calculate, by taking moments about R2, the size of the reaction R1.

(ii) Determine the size of the reaction R2.

5. The structure below is used to transport a 500kg container.



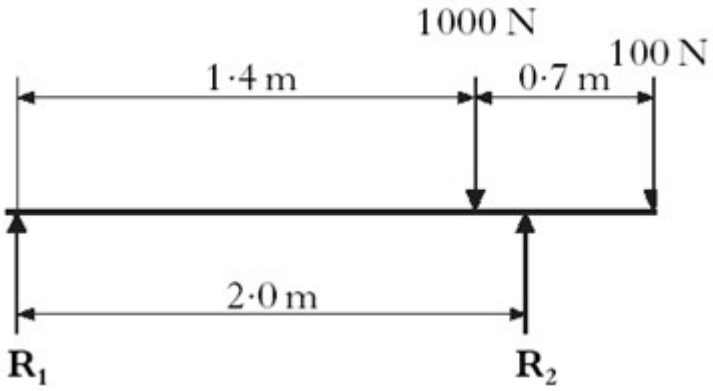
(i) Calculate the weight of the 500 kg container.

(ii) Draw the free body diagram of the structure, showing all forces and distances.

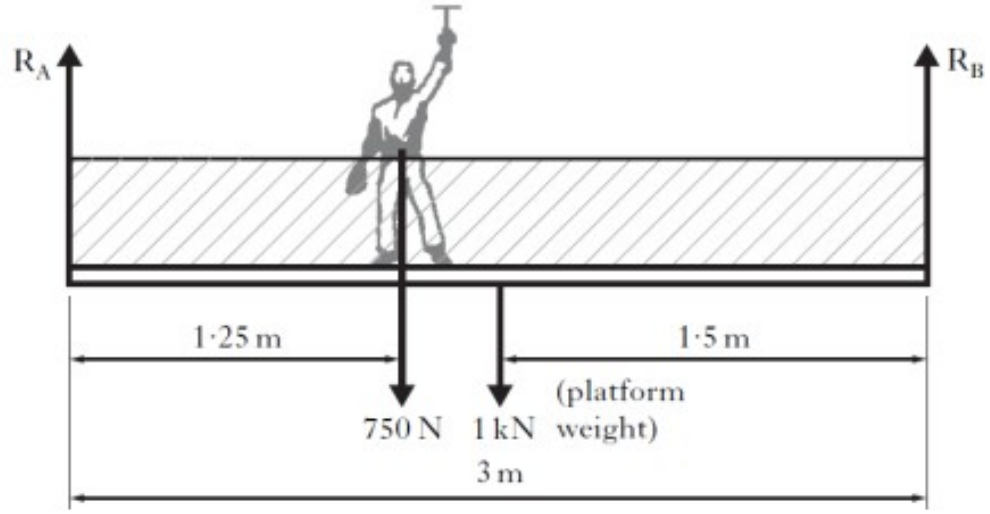
(iii) Calculate, by taking the moments about R2, the size of the reaction R1

(iv) Determine the size of the reaction R2.

6. A free body diagram for a car trailer is shown below. Taking moments about R1, calculate the reaction R2.



7. The image below shows a window cleaner's platform.

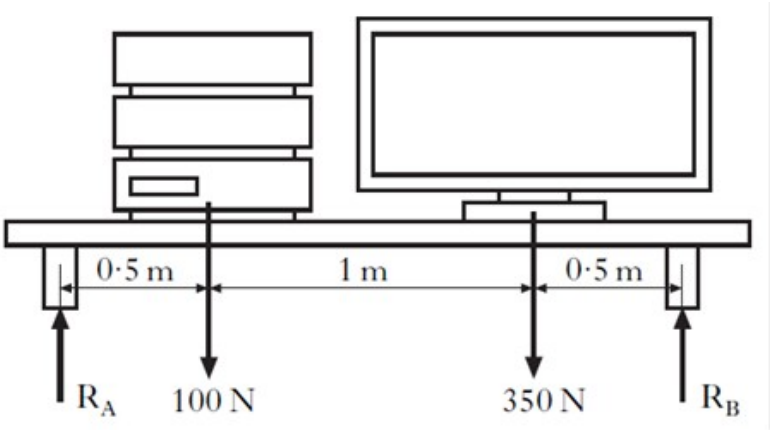


(i) Draw a free body diagram of the system.

(ii) Calculate the reaction force RA (take moments about RB)

(iii) Calculate the reaction force RB.

8. A unit used to support the DVD player, an entertainment system and television is shown.



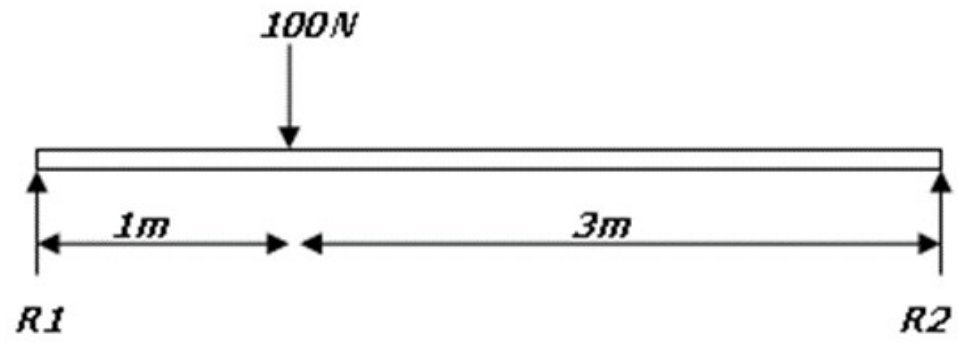
(i) Draw a free body diagram for the unit shown.

(ii) Calculate the reaction force RA (take moments about RB)

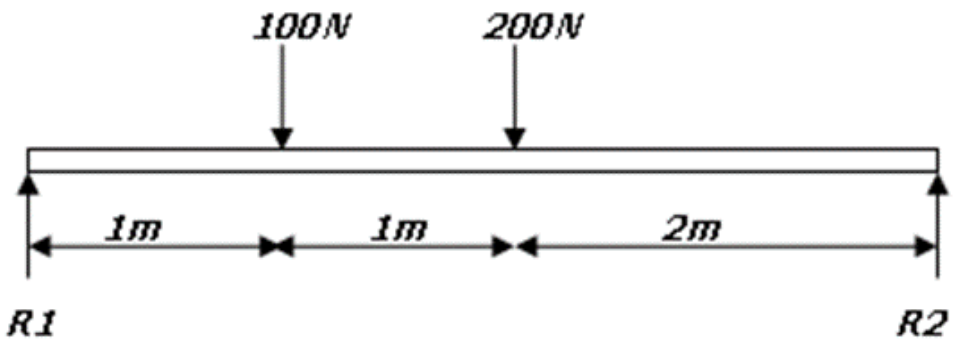
(iii) Calculate the reaction force RB.

9. Copy the free body diagrams into your jotter and determine the reactions R1 and R2 for the simply supported beams in the following examples (beam weights will be ignored in this case).

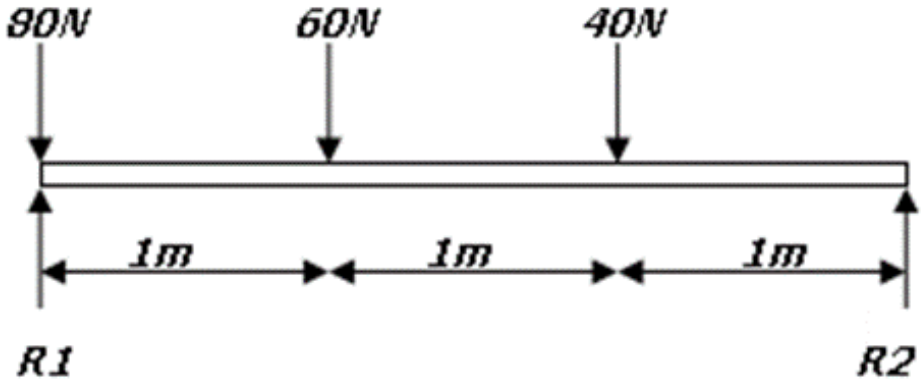
(a)



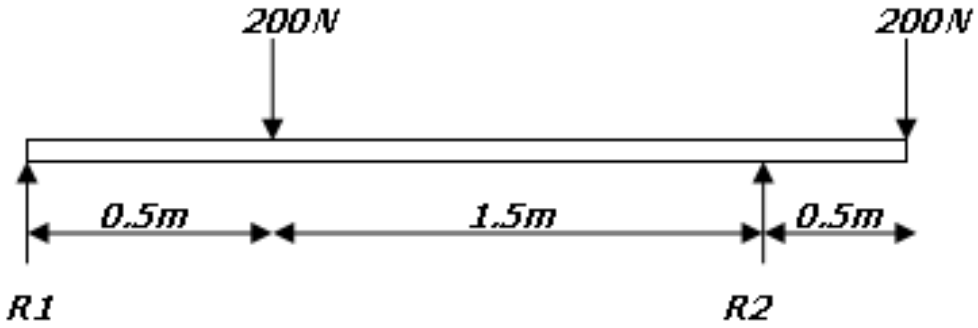
(b)



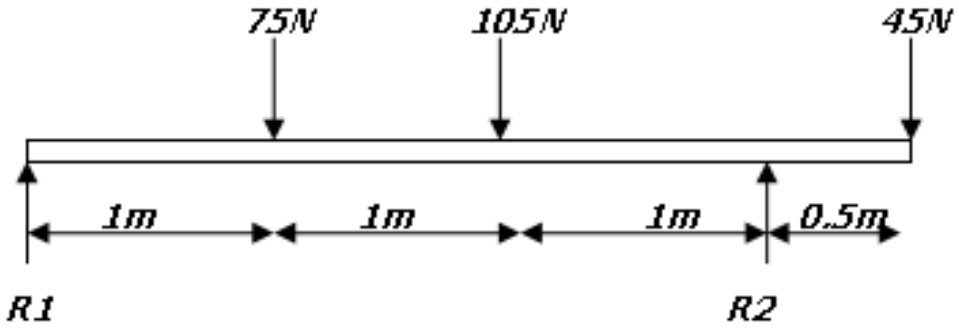
(c)



(d)

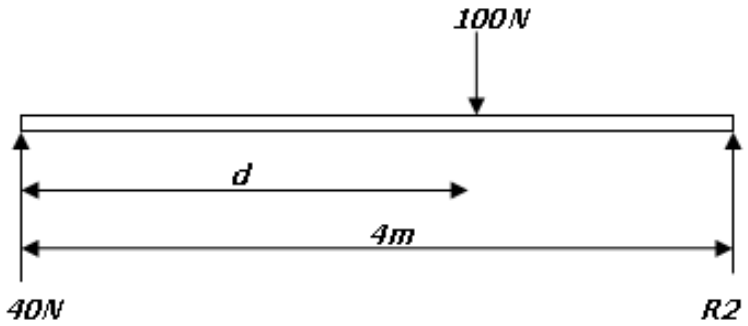


(e)

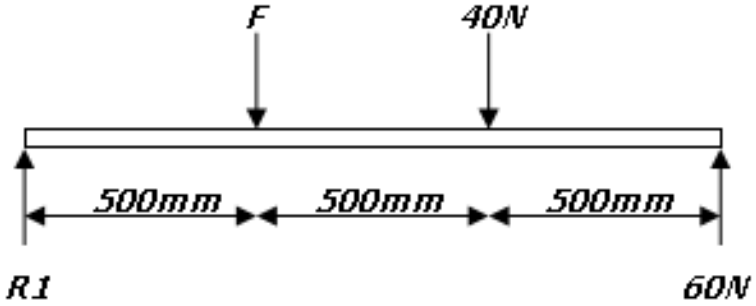


10. Copy the free body diagrams into your jotter and calculate the missing quantity. Use the principle of moments and assume that each beam is in a state of equilibrium.

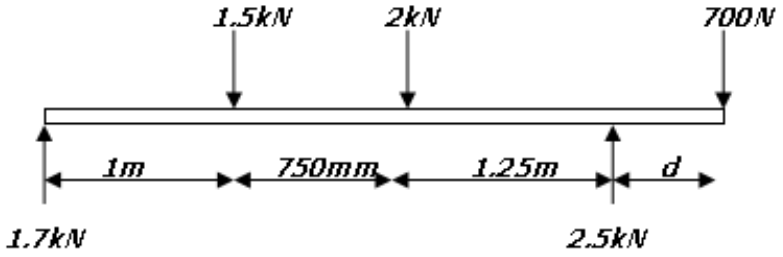
(a)


100N
d
4m
40N
R2

(b)


F
40N
500mm
500mm
500mm
R1
60N

(c)


1.5kN
2kN
700N
1m
750mm
1.25m
d
1.7kN
2.5kN